COSC 6397
Quantum Computing
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The query in Grover's Algorithm is a function that tells you the name of an element using its index.

The SAT problem given an arbitrary boolean function $f(x_1, x_2, \ldots, x_n)$ determine the values for each x_i such that $f() = 1$. This problem has not been solved in polynomial time. However, in special restricted forms the problem is solvable. Such as when the inputs are in CNF form so the input is a sequence of clauses $C_1 \wedge C_2 \wedge \cdots \wedge C_m$ where each clause consists of at most three variables. For example $C_1 = (x_2 \vee \bar{x}_3)$. There is a randomized algorithm that solves this problem in $O(4/3)^n$ time

Using Grover's algorithm this problem can be solved in $\sqrt{2^n} = 2^{n/2} = 1.41^n$. Grover's algorithm can be adapted to any problem that can be represented as searching for the result of a function.

Need to consider the lower bounds for randomized algorithms more often than deterministic algorithms when trying to show Quantum Supremacy.

A large number of randomized algorithms are just a clever form of random sampling. If the distribution is not given the assumption is uniform random sampling.

The probability of successfully selecting a the desired element from an array of n numbers is $1/n$. This makes the expected number of elements to be sampled before getting the desired elements $1/(1/n) = n$.

Probability basics. Finite sample space S and a probability function $p()$ which maps $S \to [0, 1]$. Commonly we also take advantage of a Random Variable X which maps $S \to [1, \ldots, \infty]$ if using sampling with replacement and $S \to [1, \ldots, n]$ without replacement. This allows us to ask $p(X = r) = \sum_S p(S|X(S) = r)$. $p(A) = \sum_{a \in A} p(a)$.

We often consider the expectation of a r.v. (aka the first moment) $E[X] = \sum_{x \in range(X)} xp(X = x)$.

In the random sampling problem X is the number of queries.

The most common random variables in randomized algorithms:

Bernoulli (aka 0-1 or indicator) r.v. which takes values 0 with probability $1 - p$ and 1 with probability p.

Binomial random variable $B(n, p)$ which is just a sum of indicator random variables $B(n, p) = \sum_{i=1}^{n} X_i$ where $X_i = 1$ with probability p and 0 o.w. Each trial is independent from other trials. Tells the number of successes in n trials where each succeeds with probability p. $E[B(n, p)] = \sum_{k=0}^{n} p(B(n, p) = k) = \binom{n}{k} p^k (1 - p)^{n-k} = np$. Derivation of the $E[B(n, p)]$

$E[B(n, p)] = E[\sum_{i=1}^{n} X_i] = \sum_{i=1}^{n} E[X_i] = \sum_{i=1}^{n} p = np$

Linearity of Expectation: Consider a random variables $X = \sum_{i=1}^{n} X_i$ where X_i is an arbitrary random variable then $E[X] = E[\sum_{i=1}^{n} X_i] = \sum_{i=1}^{n} E[X_i]$

For independent random variables A, B we have $p(A \cap B) = p(A)p(B)$

Mutual-Exclusion $A \cap B = \emptyset$. This is different from independence

The other most common random variables is the geometric random variable $X \sim G(p)$ which is a sequence of Bernoulli trials and determine the number of trials before getting the first success.

$E[G(p)] = 1/p$

Derivation of expectation of $G(p)$:

$E[\sum_{k=1}^{\infty} kp(G(p) = k)] = \sum_{k=1}^{\infty} k(1 - p)^{k-1} p$

Conditional Probability $p(A|B) = \frac{p(A \cap B)}{p(B)}$. Using this we can determine how often an event A occurs when we know event B has already occurred.

For R.V. X, Y we have $p(X = x)p(Y = y) = p(X = x \cap Y = y)$

using this we define the conditional expectation $E[X|Y]$. unlike the expectation which is typically a value this is a random variable which is a function of Y.

$E[X|Y] = E[X|Y] = \sum_{x \in range(X)} xp(X = x|Y)$

Taking the expectation of a conditional expectation

$E[X|Y] = E[f(Y)] = \sum_y \sum_x xp(X = x|Y = y) = E[X]$

Back to $E[G(p)]$ let X be $G(p)$. we will condition on the value of the first trial

$E[X] = E[E[X|X_1]] = E[X|X_1 = 0]p(X_1 = 0) + E[X|X_1 = 1]p(X_1 = 1)$

$= (1-p)E[X|X_1 = 0] + pE[X|X_1 = 0] = (1-p)(1-p)(E[X]+1) + p$

$= (1-p)E[X] + 1 = E[X] - pE[X] + 1$

$E[X] = E[X] - pE[X] + 1$

$pE[X] = 1$

$E[X] = 1/p$

Randomized Algorithm two classes:

Las Vegas: will always be correct (zero error),

Monte Carlo: not guaranteed to be correct (non-zero error)

Random sampling lower bound derivation:

Yao's min-max principle